

MODULE-I

Two Dimensional Concurrent Force System

Engineering Mechanics

The science which deals with the state of rest or the state of motion of the body under the action of force is called mechanics.

The application^{of} this science to actual problem is termed as Engineering Mechanics or Applied Mechanics.

Laws of Mechanics

1. First Law of Newton:

It states that, "Every body continues in the state of rest or of motion in a straight line until unless it is compelled by an external agency."

2. Second Law of Newton:

It states that, "The rate of change of momentum of a body is directly proportional to impressed force and it takes place in the direction of force applied."

$$\text{i.e. Momentum} = \text{Mass} \times \text{Velocity}$$

$$\text{Force} \propto \text{Mass} \times \text{Rate of change of velocity}$$

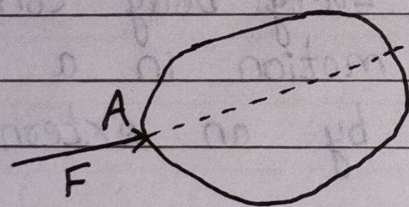
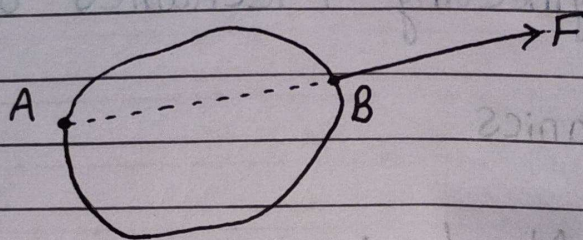
$$\text{Force} \propto \text{Mass} \times \text{Acceleration}.$$

3. Third Law of Newton:

According to this law, "Every action there is a equal and opposite reaction."

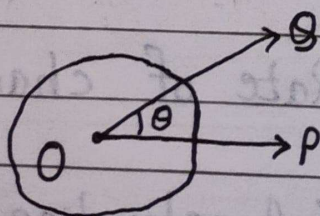
4. Law of Transmissibility of Force:

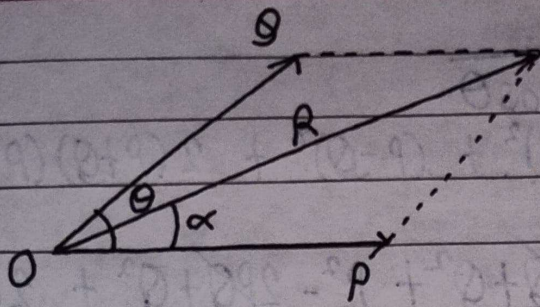
According to this law, "The state of rest or motion of the rigid body is unaltered if a force acting on a body is replaced by the another force of same magnitude and direction but acting anywhere on the body along the line of action of replaced force."



5. Law of Parallelogram:

This law states that, "If two forces acting simultaneously on a body at a point are represented in magnitude and direction by the two adjacent sides of the parallelogram, their resultant in magnitude and direction by the diagonal of the parallelogram."





$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$$

$$\tan \alpha = \frac{Q \sin \theta}{P + Q \cos \theta}$$

$$\alpha = \tan^{-1} \left(\frac{Q \sin \theta}{P + Q \cos \theta} \right)$$

Numericals

- Q) Two forces of equal magnitude of P act at an angle of θ to each other. What will be their resultant.

Solⁿ:

$$\begin{aligned} R &= \sqrt{P^2 + P^2 + 2 \times P \times P \cos \theta} \\ &= \sqrt{2P^2 + 2P^2 \cos \theta} \\ &= \sqrt{2P^2 (1 + \cos \theta)} \end{aligned}$$

$$\therefore R = P \sqrt{2 + 2 \cos \theta}$$

- Q) The resultant of the two forces $(P+Q)$ & $(P-Q)$ equals to $\sqrt{3P^2 + Q^2}$. Show that, the forces are inclined to each other at an angle of 60° .

Solⁿ:

$$R = \sqrt{P^2 + Q^2 + 2PQ \cos \theta}$$

$$\text{or, } (\sqrt{3P^2 + Q^2})^2 = (P+Q)^2 + (P-Q)^2 + 2(P+Q)(P-Q) \cos \theta$$

$$\text{or, } 3P^2 + Q^2 = P^2 + \cancel{2PQ} + Q^2 + P^2 - \cancel{2PQ} + Q^2 + 2(P+Q)(P-Q) \cos \theta$$

$$\text{or, } P^2 - Q^2 = 2(P+Q)(P-Q) \cos \theta$$

$$\text{or, } (P+Q)(P-Q) = 2(P+Q)(P-Q) \cos \theta$$

$$\text{or, } \cos \theta = 1/2$$

$$\therefore \theta = 60^\circ \text{ proved}$$

Q) The resultant of the two forces P & Q is R . If one of forces is reversed in direction then the resultant is S . Prove that: $R^2 + S^2 = 2(P^2 + Q^2)$.

Solⁿ:

$$R^2 = P^2 + Q^2 + 2PQ \cos \theta \quad \text{--- (1)}$$

And,

$$S^2 = (-P)^2 + Q^2 - 2PQ \cos \theta \quad \text{--- (2)}$$

From (1),

$$S^2 = R^2 - 2PQ \cos \theta - 2PQ \cos \theta$$

$$\text{or, } R^2 + S^2 = 4PQ \cos \theta$$

$$R^2 + S^2 = P^2 + Q^2 + 2PQ \cos \theta + P^2 + Q^2 - 2PQ \cos \theta$$

$$\Rightarrow R^2 + S^2 = 2(P^2 + Q^2) \text{ proved}$$

- 9) The resultant of the two forces, one of which is double the other, is 260 N. If the direction of the larger force is reversed and the other remains unaltered, the resultant reduces to 180 N. Determine the magnitude of the forces and the angle between the forces.

Solⁿ:

1st, $F_1 = x$

$F_2 = 2x$

$R = 260 \text{ N}$

2nd, $F_1 = x$

$F_2 = -2x$

$R = 180 \text{ N}$

$\theta = ?$

Then,

$$(260)^2 = x^2 + 4x^2 + 2 \times x \times 2x \cdot \cos \theta$$

$$\text{or, } 67600 = 5x^2 + 4x^2 \cos \theta$$

$$\text{or, } 67600 = x^2(5 + 4 \cos \theta) \text{ — (1)}$$

Also,

$$(180)^2 = x^2 + 4x^2 - 4x^2 \cos \theta$$

$$\text{or, } 32400 = 5x^2 - 4x^2 \cos \theta$$

$$\text{or, } 32400 = x^2(5 - 4 \cos \theta) \text{ — (2)}$$

Dividing eq^{ns}. (1) & (2)

$$\frac{67600}{32400} = \frac{x^2(5+4\cos\theta)}{x^2(5-4\cos\theta)}$$

$$\therefore 2.086(5-4\cos\theta) = 5+4\cos\theta$$

$$\therefore 10.43 - 8.344\cos\theta = 5+4\cos\theta$$

$$\therefore 5.43 = 12.344\cos\theta$$

$$\therefore \cos\theta = 0.4398$$

$$\therefore \theta = 64^\circ$$

Also,

$$67600 = x^2(5+4 \times 0.4398)$$

$$\therefore x^2 = \frac{67600}{6.7592} = 10001.18$$

$$\therefore x = 100 \text{ N}$$

Hence, the forces are 100 N and 200 N, and the angle between them is 64° .

Force

It is defined as an external agency ^{which} ~~with~~ changes or tends to change the condition of rest or motion of a body.

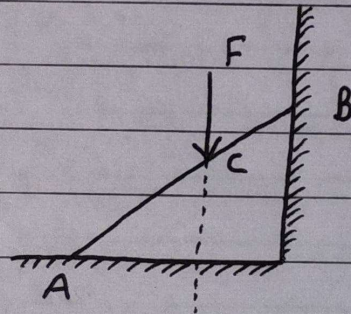
$$F = ma$$

$$F = \text{Kg} \times \frac{\text{m}}{\text{sec}^2} = \text{N}$$

$$F = mg$$

Characteristics of Force

- Magnitude of Force (F)
- Point of ~~For~~ Application (C)
- Line of Action (Vertical)
- Direction of Force (Downward)

System of Force1. Co-planar Force System:

If all the forces in a system lie in a single plane, then this is called a co-planar force system.

2. Concurrent Force System:

If the line of action of all the forces passes through a single point, called a concurrent force system.

3. Co-linear Force System:

If the line of action of all forces acts along the same line of action, called co-linear force system.

Moment of Force

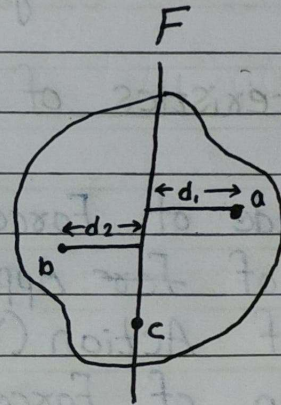
Moment is defined as the product of the magnitude of the force and perpendicular distance of the point from the line of action of force.

$$M = F \times d$$

$$M_A = F \times d_1$$

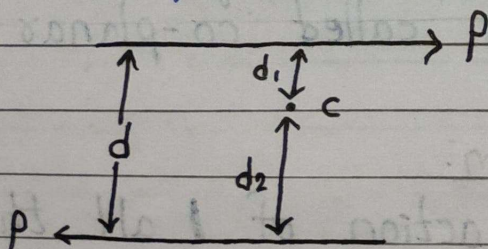
$$M_B = F \times d_2$$

$$M_C = F \times 0 = 0$$

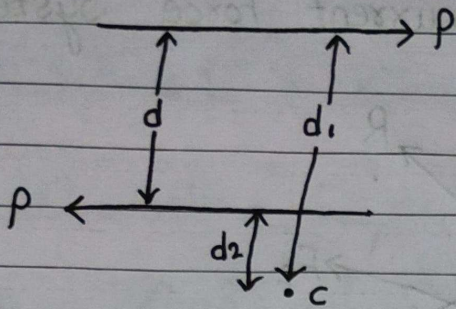


Couple

Two forces equal in magnitude and opposite in direction, are separated by a definite distance are said to form a couple.



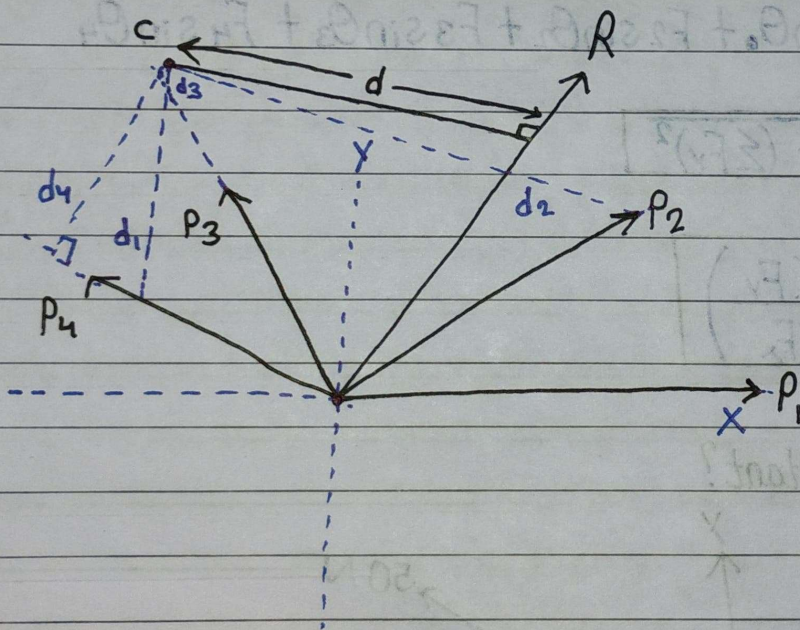
$$M_c = P \times d_1 + P \times d_2 = P \times d$$



$$M_c = P.d_1 - P.d_2 = P \times d.$$

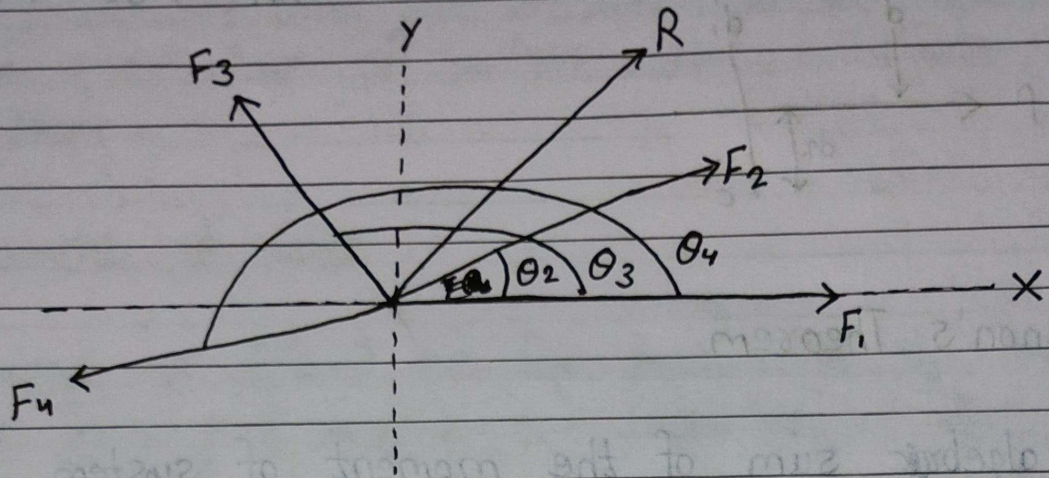
Varignon's Theorem

The algebraic sum of the moment of system about a moment centre in a co-planar concurrent force system is equal to the moment of their resultant about the moment centre.



$$P_1 d_1 + P_2 d_2 + P_3 d_3 + P_4 d_4 + \dots = R.d$$

Resultant of Co-planar Concurrent Force System



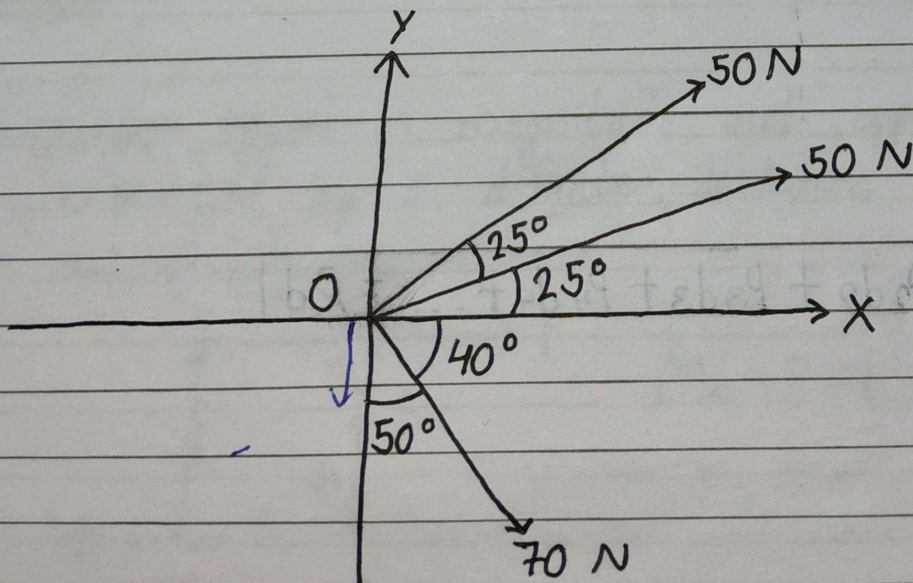
$$\sum F_x = F_1 \cos \theta_1 + F_2 \cos \theta_2 + F_3 \cos \theta_3 + F_4 \cos \theta_4$$

$$\sum F_y = F_1 \sin \theta_1 + F_2 \sin \theta_2 + F_3 \sin \theta_3 + F_4 \sin \theta_4$$

$$R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2}$$

$$\theta = \tan^{-1} \left(\frac{\sum F_y}{\sum F_x} \right)$$

Q) Find the resultant?



Solⁿ:

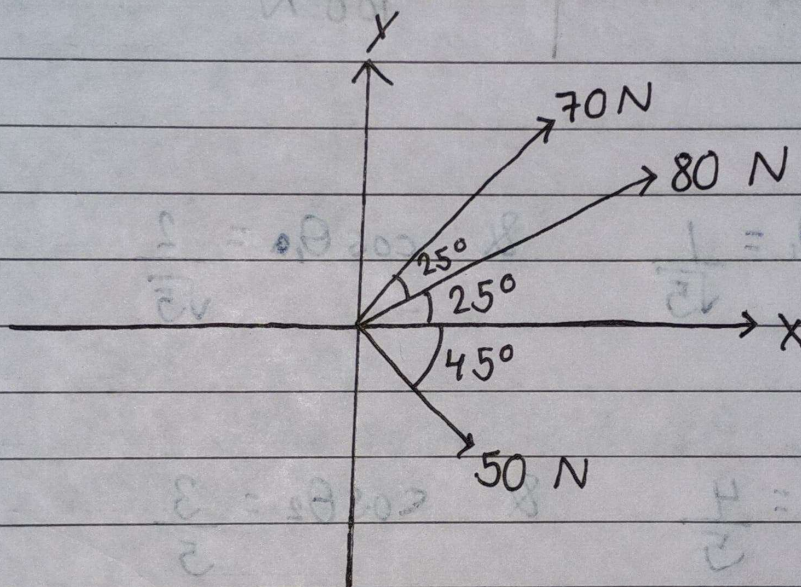
$$\sum F_x = 50 \cos 25 + 50 \cos 50 + 70 \cos(-40) = 131.077 \text{ N}$$

$$\sum F_y = 50 \sin 25 + 50 \sin 50 + 70 \sin(-40) = 14.438 \text{ N}$$

$$\therefore R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2} = 131.86 \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{\sum F_y}{\sum F_x} \right) = 6.285^\circ$$

(ii)

Solⁿ:

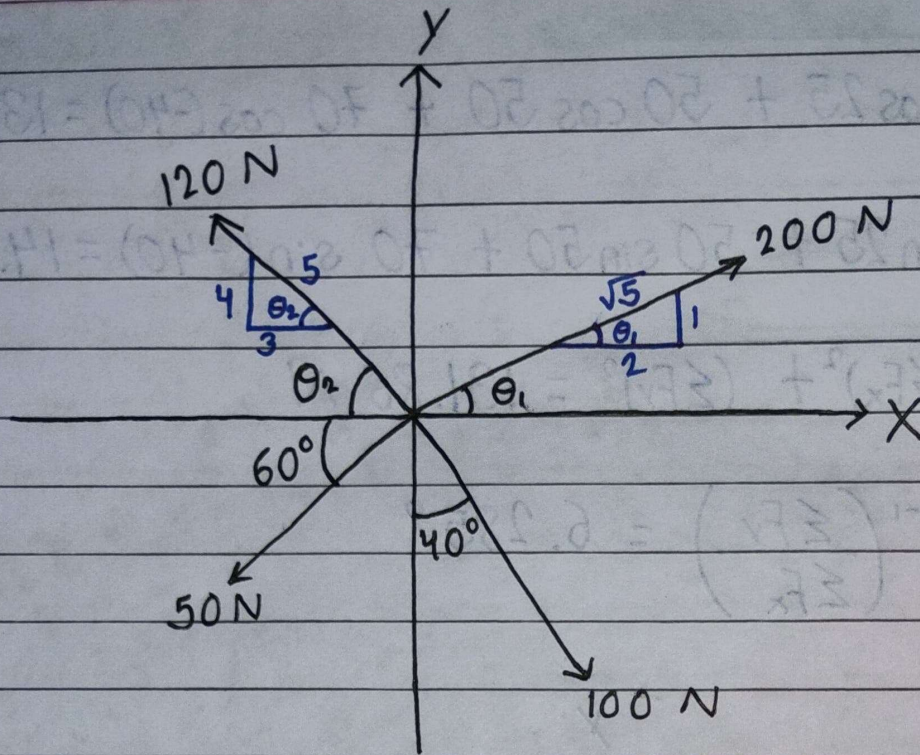
$$\sum F_x = 80 \cos 25 + 70 \cos 50 + 50 \cos 45 = 152.86 \text{ N}$$

$$\sum F_y = 80 \sin 25 + 70 \sin 50 - 50 \sin 45 = 52.07 \text{ N}$$

$$\therefore R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2} = 161.48 \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{\sum F_y}{\sum F_x} \right) = 18.81^\circ$$

(iii)

Solⁿ:For θ_1 ,

$$\sin \theta_1 = \frac{1}{\sqrt{5}} \quad \& \quad \cos \theta_1 = \frac{2}{\sqrt{5}}$$

For θ_2 ,

$$\sin \theta_2 = \frac{4}{5} \quad \& \quad \cos \theta_2 = \frac{3}{5}$$

Now,

$$\begin{aligned} \sum F_x &= 200 \cos \theta_1 - 120 \cos \theta_2 - 50 \cos 60 + 100 \cos 50 \\ &= 146.17 \text{ N} \end{aligned}$$

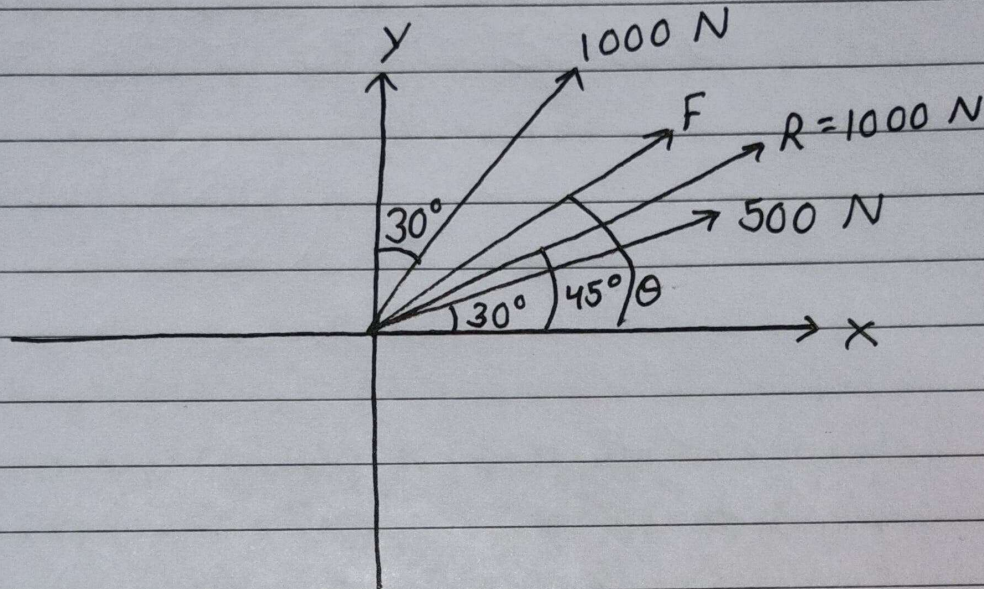
$$\begin{aligned} \sum F_y &= 200 \sin \theta_1 + 120 \sin \theta_2 - 50 \sin 60 - 100 \sin 50 \\ &= 65.54 \text{ N} \end{aligned}$$

$$\therefore R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2} = 160.19 \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{\sum F_y}{\sum F_x} \right) = 24.15^\circ$$

- Q) Two forces acting on a body are 500 N and 1000 N . Determine the third force ' F ', such that the resultant of all three forces is 1000 N directed at 45° to X-axis.

Solⁿ:



Given,

$$\sum F_x = 1000 \cos 45$$

$$\therefore 500 \cos 30 + 1000 \cos 60 + F \cos \theta = 1000 \cos 45$$

$$\therefore F \cos \theta = 225.90 \text{ --- (1)}$$

$$\sum F_y = 1000 \sin 45$$

$$\therefore 500 \sin 30 + 1000 \sin 60 + F \sin \theta = 1000 \sin 45$$

$$\therefore F \sin \theta = 408.91 \text{ --- (2)}$$

Dividing (2) by (1),

$$\frac{\sin \theta}{\cos \theta} = \frac{408.91}{225.90} = 1.8101$$

$$\therefore \theta = \tan^{-1}(1.8101) = 61.08^\circ$$

$$\& F = 467.13\text{ N}$$

Ans